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/**
 * QuantumCap Engine Framework
 * Architecture Focus: Clean State Management, Fetch API error resilience,
DOM mutations
 */

// Global Application Core State
const State = {
  assets: [],
  filteredAssets: [],
  searchQuery: '',
  isFetching: false
};

// DOM Node Cache Map
const DOM = {
  tableBody: document.getElementById('asset-table-body'),
  searchInput: document.getElementById('search-input'),
  refreshBtn: document.getElementById('refresh-btn'),
  refreshIcon: document.getElementById('refresh-icon'),
  errorContainer: document.getElementById('error-container'),
  errorMessage: document.getElementById('error-message'),
  totalMarketCap: document.getElementById('total-market-cap'),
  topGainer: document.getElementById('top-gainer'),
  topGainerVal: document.getElementById('top-gainer-val'),
  connectionStatus: document.getElementById('connection-status')
};

// REST API endpoint configuration (CoinGecko public API)
const API_ENDPOINT = 'https://api.coingecko.com/api/v3/coins/markets?
vs_currency=usd&order=market_cap_desc&per_page=15&page=1&sparkline=f
alse';

/**
 * Initialize Dashboard Application Context
 */
document.addEventListener('DOMContentLoaded', () => {
  // Structural layout icons initialization via Lucide agent
  if (window.lucide) window.lucide.createIcons();

  // Bind Interface Event Listeners
  DOM.refreshBtn.addEventListener('click', fetchMarketData);
  DOM.searchInput.addEventListener('input', handleSearch);

  // Kickoff Initial HTTP Network Request
  fetchMarketData();
});

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/**
 * Orchestrate async HTTP network lifecycle routine
 */
async function fetchMarketData() {
  if (State.isFetching) return;

  setLoadingState(true);
  hideErrorBanner();

  try {
    const response = await fetch(API_ENDPOINT);

    if (!response.ok) {
      throw new Error(`HTTP network fault layer responded with status code:
${response.status}`);
    }

    const data = await response.json();

    // Mutate System State
    State.assets = data;
    State.filteredAssets = applySearchFilter(data, State.searchQuery);

    // Render Updated State Streams
    renderGlobalMetrics();
    renderTableData();

  } catch (error) {
    console.error('Pipeline Fault Detected:', error);
    displayErrorBanner(error.message || 'Transient error during network
handshake lifecycle execution.');
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  } finally {
    setLoadingState(false);
  }
}

/**
 * Compute and render premium high-level metrics cards
 */
function renderGlobalMetrics() {
  if (!State.assets.length) return;

  // 1. Calculate Aggregate Evaluated Portfolio Market Cap
  const absoluteCap = State.assets.reduce((accumulator, item) =>
accumulator + (item.market_cap || 0), 0);
  DOM.totalMarketCap.textContent = new Intl.NumberFormat('en-US', { style:

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'currency', currency: 'USD', maximumFractionDigits: 0 }).format(absoluteCap);
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// 2. Compute Array Matrix Max Delta to isolate top asset performer
const topAsset = [...State.assets].sort((alpha, beta) =>
beta.price_change_percentage_24h - alpha.price_change_percentage_24h)
[0];
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if (topAsset) {
  DOM.topGainer.textContent = topAsset.name;
  DOM.topGainerVal.textContent = `+$
{topAsset.price_change_percentage_24h.toFixed(2)}%`;
}
}
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/**
 * Construct and write clean table row node components inside active UI frame
 */
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function renderTableData() {
  DOM.tableBody.innerHTML = '';

  if (State.filteredAssets.length === 0) {
    DOM.tableBody.innerHTML = `
      <tr>
        <td colspan="5" class="py-12 px-6 text-center text-slate-500 font-
medium">
          No active tokens matched your query filter parameter state.
        </td>
      </tr>`;
    return;
  }
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```
State.filteredAssets.forEach(token => {
  const rowNode = document.createElement('tr');
  rowNode.className = 'hover:bg-slate-800/30 transition-colors
duration-150 group border-b border-slate-800/40 last:border-0';

  const deltasNegative = token.price_change_percentage_24h < 0;
  const formattedPrice = new Intl.NumberFormat('en-US', { style: 'currency',
currency: 'USD', minimumFractionDigits: 2 }).format(token.current_price);
  const formattedCap = new Intl.NumberFormat('en-US', { style: 'currency',
currency: 'USD', maximumFractionDigits: 0 }).format(token.market_cap);
  const formattedVolume = new Intl.NumberFormat('en-US', { style:
'currency', currency: 'USD', maximumFractionDigits:
0 }).format(token.total_volume);
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rowNode.innerHTML = `
  <td class="py-4 px-6 flex items-center gap-3">
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        <div>
            <span class="block font-semibold text-slate-200 group-hover:text-
indigo-400 transition-colors">${token.name}</span>
            <span class="block text-xs text-slate-500 font-mono
uppercase">${token.symbol}</span>
        </div>
    </td>
    <td class="py-4 px-6 text-right font-mono font-medium text-
slate-200">
        ${formattedPrice}
    </td>
    <td class="py-4 px-6 text-right font-mono font-semibold $
{deltalsNegative ? 'text-rose-400' : 'text-emerald-400'}">
        ${deltalsNegative ? '' : '+'}$
        {token.price_change_percentage_24h?.toFixed(2)}%
    </td>
    <td class="py-4 px-6 text-right font-mono text-slate-400 hidden
md:table-cell">
        ${formattedCap}
    </td>
    <td class="py-4 px-6 text-right font-mono text-slate-400 hidden
sm:table-cell">
        ${formattedVolume}
    </td>
    `;
    DOM.tableBody.appendChild(rowNode);
  });
}

/**
 * Handle Search Input Matrix Filtering
 */
function handleSearch(event) {
  State.searchQuery = event.target.value.toLowerCase().trim();
  State.filteredAssets = applySearchFilter(State.assets, State.searchQuery);
  renderTableData();
}

function applySearchFilter(collection, query) {
  if (!query) return collection;
  return collection.filter(element =>
    element.name.toLowerCase().includes(query) ||
    element.symbol.toLowerCase().includes(query)
  );
}

```

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/**
 * Toggle Component UI Loading Presentation Wrapper States
 */
function setLoadingState(loading) {
  State.isFetching = loading;
  if (loading) {
    DOM.refreshBtn.disabled = true;
    DOM.refreshIcon.classList.add('animate-spin');
    DOM.connectionStatus.className = "inline-flex items-center gap-1.5
px-2.5 py-0.5 rounded-full text-xs font-medium bg-amber-500/10 text-
amber-400 border border-amber-500/20";
    DOM.connectionStatus.innerHTML = `<span class="w-1.5 h-1.5 rounded-
full bg-amber-400 animate-ping"></span> Querying API...`;
  } else {
    DOM.refreshBtn.disabled = false;
    DOM.refreshIcon.classList.remove('animate-spin');
    DOM.connectionStatus.className = "inline-flex items-center gap-1.5
px-2.5 py-0.5 rounded-full text-xs font-medium bg-emerald-500/10 text-
emerald-400 border border-emerald-500/20";
    DOM.connectionStatus.innerHTML = `<span class="w-1.5 h-1.5 rounded-
full bg-emerald-400 animate-pulse"></span> Live Data Feed`;
  }
}

function displayErrorBanner(message) {
  DOM.errorMessage.textContent = message;
  DOM.errorContainer.classList.remove('hidden');
}

function hideErrorBanner() {
  DOM.errorContainer.classList.add('hidden');
}

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